Identity, Scarcity, and the Distribution of Resources: An Agent-Based Model of Group-Structured Cooperation

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August 2026

Abstract

Ethnically diverse societies frequently display both lower cooperation and persistent between-group inequality, and it is tempting to read the second as a consequence of the first. We build an agent-based model in which the probability of cooperating depends jointly on ethnic, cultural and social distance, on learned reputation at the group and individual level, and on the actor's own economic position, and in which the resulting payoffs feed back into the resource distribution. Agents play a symmetric donation game and an asymmetric trust game on an endogenously rewiring homophilous network. Because each channel can be disabled independently, their contributions can be separated rather than asserted.

Four results stand out. First, symmetric in-group favouritism on its own generates almost no between-group inequality: holding identity bias at its baseline level but equalising access to returns collapses the between-group share of total inequality from 22.3 % to 1.5 %. Differential access to returns, not differential treatment, is what opens a durable group gap. Second, taste-based preference and network homophily are near-orthogonal channels: removing the first cuts the in-group cooperation bias from 0.52 to 0.18 while leaving network segregation almost intact, and removing the second collapses segregation from 0.62 to 0.14 while leaving the bias almost intact. Third, group-level reputation learning does not manufacture bias where none exists, but it does convert a purely economic disadvantage into a group-level stigma: with *zero* taste for discrimination, raising the learning weight alone raises the destitute share from 5.4 % to 14.8 %. Fourth, economic stress *dissolves* rather than sharpens the in-group premium — as scarcity sensitivity and shock exposure rise together, in-group bias falls from 0.58 to 0.28, because desperate agents defect on co-ethnics too.

The model is deliberately a caricature and its magnitudes are properties of one parameterisation; the counterfactual directions, which survive across seeds and sweeps, are the portable part. Code, tests and an interactive results viewer are provided.

1 Introduction

Two empirical regularities motivate this paper. The first is that ethnically diverse communities tend to under-provide public goods, a pattern documented across US municipalities (Alesina et al., 1999), Kenyan school committees (Miguel and Gugerty, 2005), and Ugandan urban neighbourhoods (Habyarimana et al., 2007). The second is that resources in such societies are often distributed persistently unevenly *along* group lines, with minority or historically disadvantaged groups holding far less than their population share.

The intuitive bridge between the two is that people cooperate more readily with their own kind. That in-group favouritism is real and robust: it appears in minimal-group experiments (Tajfel et al., 1971), in third-party punishment (Bernhard et al., 2006b,a), in segmented-society trust games (Fershtman and Gneezy, 2001), and it survives meta-analysis (Balliet et al., 2014). Combined with the strong homophily

of real social networks (McPherson et al., 2001; Currarini et al., 2009), it is easy to tell a story in which favouritism plus segregation mechanically produce group-level economic gaps.

This paper asks whether that story is actually true in a model built to give it every chance. We construct an agent-based economy in which the probability of cooperation depends on ethnic, cultural and social distance, on learned reputation at both the group and the dyad level, and on the agent's own economic position; the resulting payoffs then move resources around, and the new distribution feeds back into who is willing to cooperate with whom tomorrow. Because every channel is a separate parameter, we can switch each off and read its contribution.

The headline finding is negative, and it is the reason the paper exists. *Symmetric in-group favouritism, however strong, barely moves the between-group distribution.* If every group favours its own equally, the favouritism roughly cancels across groups; what remains is a small majority advantage from the fact that a member of a large group meets co-ethnics more often. In our baseline that residual is worth a few percentage points of resource share, against a between-group gap that is an order of magnitude larger and traceable almost entirely to differential access to returns.

That does not make identity irrelevant. It relocates its role. Identity bias governs the *texture* of cooperation — who trades with whom, how segregated the interaction network becomes — and, through group-level reputation learning, it provides the machinery by which an economic disadvantage is converted into a durable group-level stigma. The interesting interaction in the model is not identity $\times$ inequality but *stereotype learning* $\times$ inequality: agents who generalise a poor partner's behaviour to that partner's whole group end up excluding a disadvantaged group even when they harbour no taste for discrimination at all. That is the mechanism of statistical discrimination (Phelps, 1972; Arrow, 1973; Coate and Loury, 1993) arising endogenously from a distributional shock.

The paper's contributions are: (i) a model that combines identity-conditioned cooperation, endogenous network segregation, reinforcement learning at two levels, and a resource process capable of poverty traps, in one framework; (ii) a clean decomposition of five mechanisms via matched counterfactuals; and (iii) two two-dimensional parameter sweeps that locate where the mechanisms interact and where they do not. Section 2 specifies the model, Section 3 the experiments, Section 4 the results, and Sections 5–6 discuss and delimit them.

2 Model

The model runs in discrete rounds. Each round, agents earn income, are matched into pairs, play one of two games, update beliefs and possibly ties, and consume. We describe each component in turn; Table 2 lists the baseline parameter values.

2.1 Agents

The population contains N agents partitioned into G ethnic groups with population shares $(\pi_1, \ldots, \pi_G)$. Group membership is fixed. Agent i carries:

- **Ethnicity** $e_i \in \{1, \ldots, G\}$.
- **Culture** $c_i \in [0,1]^K$, drawn around a group-specific centroid with within-group dispersion σ_c. Dimension c_{i1} is a *cooperative-norm* trait that enters the decision rule directly; the rest serve only as a similarity metric. Traits drift toward those of partners after mutually cooperative interactions.
- **Resources** $w_i \geq 0$, initialised lognormally around a group-specific mean.

- **Productivity** $a_i > 0$, a multiplier on income capturing structural access to returns — land, credit, schooling, licences — as distinct from the stock of resources currently held. It is drawn lognormally around a group-specific mean and is fixed thereafter.
- **Group beliefs** $B_i \in [-1,1]^G$, a learned valence toward each ethnic group. This is the stereotype channel.
- **Dyad beliefs** $D_{ij} \in [-1,1]$, a learned valence toward specific past partners. This is the direct-reciprocity channel.

Separating a_i from w_i matters. If groups differ only in their starting stock, the mean-reverting economy of Section 2.5 erases the difference within roughly a hundred rounds and no group gap survives to be explained. Persistent between-group inequality in this model requires persistent differences in the *rate* at which resources are generated, not merely in the initial endowment — which is itself a substantive claim, and one consistent with asset-based accounts of persistent poverty (Carter and Barrett, 2006; Barrett et al., 2016).

2.2 Interaction structure

Ties form with log-affinity decreasing in ethnic difference, cultural distance and log-resource distance, scaled so that the expected degree matches a target $\bar{d}$. This is a standard homophilous attachment kernel (McPherson et al., 2001) with three homophily weights h_e, h_c, h_w.

Each round agents are matched into pairs. With probability p_{loc} an agent seeks a partner among its unmatched network neighbours; otherwise it draws from the unmatched population at large. Ties are then revised: a fraction ρ of agents drop the neighbour who behaved worst toward them in that round and form a new tie, choosing among sampled candidates by the same homophilous affinity. Segregation is therefore an outcome, not a fixed input — it responds to behaviour, in the spirit of Schelling (1971).

2.3 The cooperation rule

Agent i's propensity to cooperate with j is a logistic function of a latent index combining structural drivers with two learned terms:

$$\begin{aligned} z_{ij} \;=\; & \beta_0 + \beta_{\text{eth}}\big(\mathbb{1}[e_i = e_j] - \bar{\pi}\big) + \beta_{\text{cult}}\left(1 - \frac{\|c_i - c_j\|}{\sqrt{K}} - \tfrac{1}{2}\right) + \beta_{\text{soc}}\, s_{ij} \\ & + \beta_{\text{norm}}\left(c_{i1} - \tfrac{1}{2}\right) + \beta_{\text{own}} \tanh\!\left(\tfrac{w_i - \bar{w}}{\bar{w}}\right) + \beta_{\text{gap}} \tanh\!\left(\tfrac{w_j - w_i}{\bar{w}}\right) + \beta_{\text{scar}}\, \kappa_i \\ & + \lambda_{\text{grp}}\, B_i[e_j] + \lambda_{\text{dyad}}\, D_{ij} + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \end{aligned} \tag{1}$$

and $P(\text{cooperate}) = \sigma(z_{ij}) = (1 + e^{-z_{ij}})^{-1}$. Here s_{ij} is social proximity (a tie indicator plus the Jaccard overlap of the two neighbourhoods), $\bar{w}$ is mean resources, and

$$\kappa_i \;=\; \max\!\left(0,\; 1 - \frac{w_i}{\eta\, \bar{s}}\right) \in [0,1] \tag{2}$$

is *scarcity*: how thin agent i's buffer is relative to η rounds of subsistence $\bar{s}$. With $\beta_{\text{scar}} < 0$, an agent close to destitution is pushed toward defection.

Why the ethnic term is centred. The indicator $\mathbb{1}[e_i = e_j]$ is centred on $\bar{\pi} = \sum_g \pi_g^2$, the probability that two randomly drawn agents share an ethnicity. This is not cosmetic. Without centring, raising β_{eth} also raises the population-average propensity to cooperate, so a "no ethnic bias" counterfactual is silently also a "less cooperation overall" counterfactual and the two effects cannot be told apart. In an early uncentred version of this model, switching off ethnic bias appeared to *raise* inequality — an artefact entirely driven by the collapse in total cooperation that came with it. Centred, β_{eth} controls the in-group/out-group *gap* and leaves the *level* alone, which is what a comparative-static exercise requires.

Learning. After each interaction, agent i observes a behavioural signal $s \in [-1, 1]$ from j and updates both belief systems by the Rescorla–Wagner rule (Rescorla and Wagner, 1972), the same error-correcting form used in reinforcement models of learning in games (Roth and Erev, 1995):

$$B_i[e_j] \leftarrow B_i[e_j] + \alpha_{\text{grp}}\big(s - B_i[e_j]\big), \qquad D_{ij} \leftarrow D_{ij} + \alpha_{\text{dyad}}\big(s - D_{ij}\big). \tag{3}$$

We set $\alpha_{\text{grp}} \ll \alpha_{\text{dyad}}$, so first-hand impressions of an individual move fast while beliefs about a whole group accumulate slowly and are correspondingly harder to dislodge. Dyadic memory decays slowly between encounters. The group channel is a deliberately crude model of the process by which behaviour is generalised to a category (Yamagishi and Kiyonari, 2000), and it is what allows a stereotype to become self-confirming in the sense of Coate and Loury (1993).

2.4 Games

A share ϕ of interactions are trust games; the remainder are donation games.

Donation game (symmetric). Both agents simultaneously choose to cooperate or defect with the probabilities given by (1). A cooperator pays a cost γ and the partner receives $m\gamma$ with $m > 1$; mutual cooperation therefore yields $(m-1)\gamma$ each. Crucially, *an agent cannot pay a cost it cannot afford*: the cost is capped at current holdings. Poverty is a hard budget constraint on cooperation, not merely a disincentive — an assumption that distinguishes this from the standard donation game (Nowak, 2006).

Trust game (asymmetric). Following Berg et al. (1995), one agent is investor and the other trustee. The wealthier agent takes the investor role with probability p_{rich}, making this the patron–client channel. The investor stakes $\sigma(z_{\text{inv}}) \cdot \tau \cdot \min(w_{\text{inv}}, \bar{\tau})$, the stake is multiplied by m_τ, and the trustee returns a share equal to its own cooperation probability. The absolute cap $\bar{\tau}$ keeps per-interaction surplus bounded; without it the economy compounds without limit (Section 2.7).

Observed signals are $s = \pm 1$ for cooperation and defection in the donation game, and $s = 2x - 1$ in the trust game where x is the share of the endowment staked or the share of the pot returned.

2.5 Resource dynamics

Income is Solow-like, with a flat component and a diminishing return on holdings, scaled by the agent's productivity and hit by a multiplicative shock:

$$y_i \;=\; a_i\big(y_0 + r\,w_i^{\theta}\big)\,\xi_i, \qquad \theta < 1, \quad \xi_i \sim \text{Lognormal}(-\tfrac{1}{2}\sigma_y^2, \sigma_y^2). \tag{4}$$

With probability q an agent suffers a *crisis* destroying a fraction λ of its holdings — illness, crop failure, expropriation — in the spirit of the shock-and-asset-dynamics literature (Rosenzweig and Wolpin, 1993;

Dercon, 2002; Carter et al., 2007). Resources then evolve as

$$w_i \leftarrow (1-\delta)\Big[w_i + y_i - (\text{crisis loss}) + \Delta_i - \bar{s}\Big]^{+}, \tag{5}$$

where Δ_i is the round's game payoff and $\bar{s}$ is subsistence. An optional flat tax t on income, redistributed as an equal per-capita transfer, provides the policy lever.

Two features of (4)–(5) do the structural work. Because $\theta < 1$ and $\delta > 0$, an isolated agent converges to a productivity-specific steady state, so heterogeneity in a_i — not heterogeneity in initial w_i — is what sustains long-run inequality. And because $a_i y_0 < \bar{s}$ for most agents, income at zero holdings falls short of subsistence: an agent knocked to the floor *cannot climb out unaided*. Escape requires being cooperated with, which is precisely what (2) makes unlikely. This is the model's poverty trap, and it is a behavioural rather than a technological one (Bowles et al., 2006; Carter and Barrett, 2006).

2.6 Measurement

We track the Gini coefficient of holdings and the Theil T index, which we decompose additively into between- and within-group components (Theil, 1967; Bourguignon, 1979; Shorrocks, 1980):

$$T = \underbrace{\sum_g \omega_g \ln \frac{\mu_g}{\mu}}_{T_{\text{between}}} + \underbrace{\sum_g \omega_g T_g}_{T_{\text{within}}}, \qquad \omega_g = \frac{n_g \mu_g}{N\mu}, \tag{6}$$

and report T_{between}/T as the *between-group share*. We also record cooperation rates split by whether the pair is co-ethnic (their difference is the *in-group bias*), the full $G \times G$ matrix of group-toward-group cooperation, Newman's assortativity coefficient of the tie network by ethnicity (Newman, 2003), the destitute share, and rank persistence of holdings against 100 rounds earlier.

2.7 Calibration, and two ways it went wrong

Calibration is load-bearing here in a way worth reporting, because both failure modes silently produce a model that looks fine and answers nothing.

In the first parameterisation the trust-game stake was an unbounded fraction of holdings, so the economy compounded at roughly 4 % per round. Within two hundred rounds every agent was far above subsistence, the scarcity term (2) was identically zero for the entire population, and the poverty channel — one of the paper's objects of study — was dead code that nonetheless ran without error.

Over-correcting produced the opposite failure. With sharply diminishing returns and no productivity heterogeneity, the economy has a *single* global attractor: every agent converges to the same steady state, the Gini falls to 0.07, and the between-group share falls to 2 % regardless of any identity parameter. The model was then incapable of exhibiting the phenomenon it was built to study.

The reported parameterisation adds heterogeneous productivity (a distribution of steady states rather than one) and crisis shocks (a mechanism that pushes agents into the region where scarcity binds). We report this because "the scarcity term is inert at these parameters" is a real and easily-missed failure mode in models of this kind, and because it establishes that the scarcity results below are not artefacts of a term that never activates.

3 Experimental design

All results use $N = 500$ agents, $G = 3$ groups with population shares $(0.5, 0.3, 0.2)$, and 350 rounds, averaged over 5 random seeds. Groups differ in mean initial resources $(12, 10, 8)$ and in mean productivity

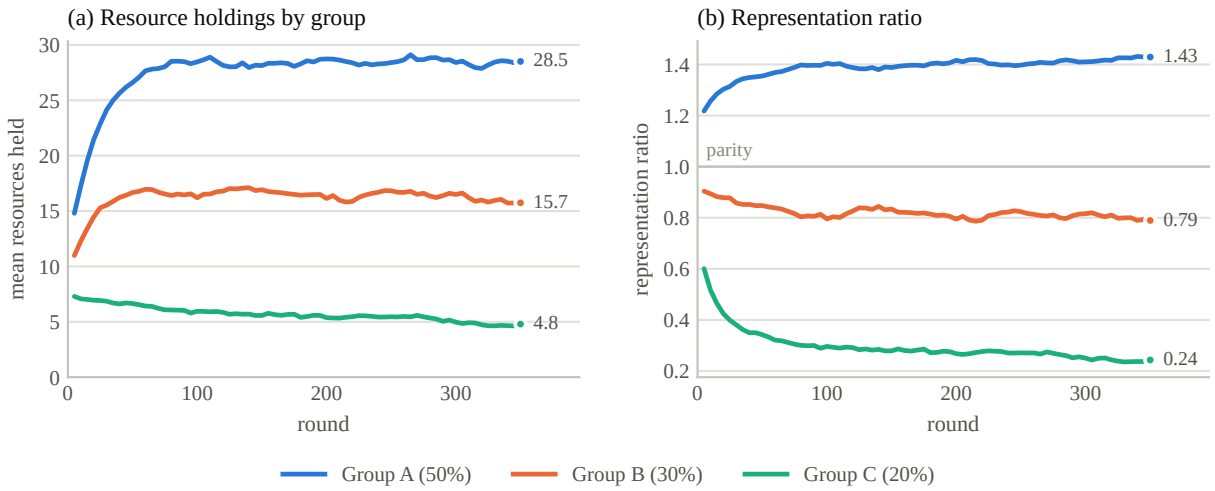


Figure 1: Baseline trajectories, mean of 5 seeds. (a) Mean resources held by group. (b) Resource share divided by population share; 1.0 is parity. Group C, a fifth of the population, converges to roughly a quarter of parity.

$(1.2, 1.0, 0.75)$. We refer to them as groups A, B and C. Reported statistics are means over the final eight recorded snapshots; across-seed standard deviations are small relative to the effects discussed (for the baseline: 0.016 for the Gini, 0.007 for in-group bias, 0.018 for assortativity).

Counterfactual scenarios. Each scenario disables exactly one mechanism relative to the baseline, except where noted: *no taste-based bias* ($\beta_{\text{eth}} = 0$); *no ethnic homophily* ($h_e = 0$, so ties ignore ethnicity but preferences do not); *no stereotype learning* ($\lambda_{\text{grp}} = 0$); *no desperation* ($\beta_{\text{scar}} = 0$); *equal access* (identical productivity and initial resources across groups, identity bias fully intact); *equal access + no bias* (both, the null world); *redistribution* (a 35 % flat tax); and *forced contact* ($p_{\text{loc}} = 0.2$ with rewiring off, so most interactions are with strangers).

Sweeps. We run two 2-D grids with 3 seeds per cell: $\beta_{\text{eth}} \in \{0, 0.6, 1.2, 1.8, 2.4\}$ crossed with $\lambda_{\text{grp}} \in \{0, 0.75, 1.5, 2.25\}$; and $\beta_{\text{scar}} \in \{0, -1.5, -3, -4.5\}$ crossed with crisis probability $q \in \{0, 0.02, 0.05, 0.09\}$.

4 Results

4.1 The baseline world

Figure 1 shows the baseline trajectories. The distribution stabilises within roughly a hundred rounds and then holds: group A settles at 28.5 mean resources, group B at 15.7, and group C at 4.8. In representation terms, group A ends holding 1.43 times its population share while group C holds 0.24 of its own — that is, 20 % of the population holding 4.9 % of the resources. The overall Gini is 0.578 and the between-group share of Theil inequality is 22.3 % (Figure 2a).

The distributional story is inseparable from a behavioural one. Figure 3a gives the cooperation matrix. Group A cooperates with its own at intensity 0.84; group C cooperates with its own at only 0.41, and with groups A and B at 0.15 and 0.12. Group C is not merely disliked — it has largely stopped cooperating with *anyone*, including itself. By the end of the run 30.2 % of its members hold nothing at all, against 0.7 % in group A (Figure 2b).

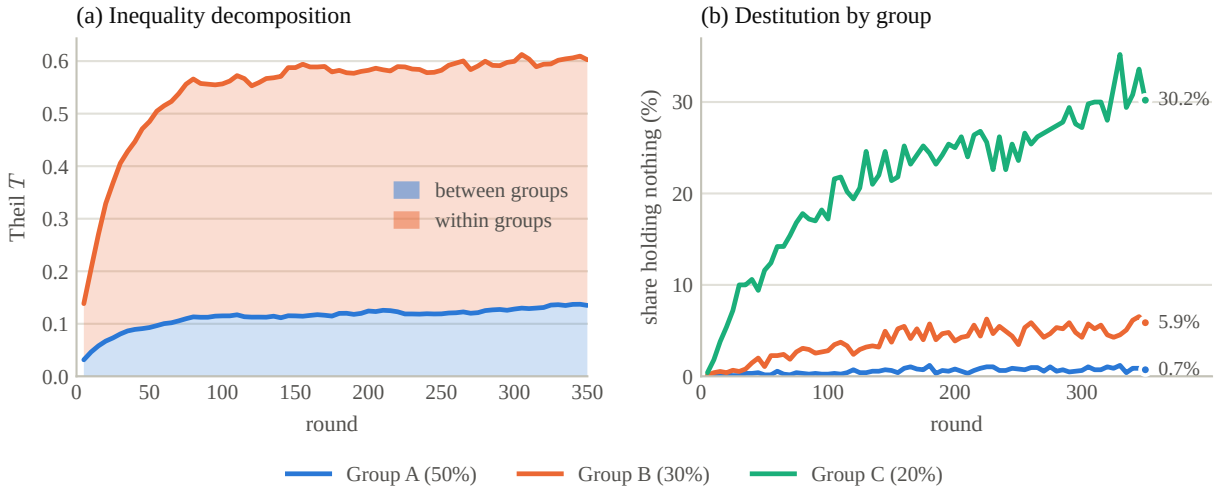


Figure 2: (a) Theil T decomposed into between- and within-group components; the between-group band is the part attributable to group membership. (b) Share of each group holding nothing. Destitution is heavily concentrated in group C and still rising at the end of the run.

Table 1: Counterfactual scenarios. Each row disables one mechanism. Values are means over 5 seeds and the final eight snapshots.

Scenario	Gini	Between-group share	Coop. rate	In-group bias	Destitute share	Assortativity
Baseline	0.578	0.223	0.518	0.522	0.082	0.620
No taste-based bias	0.573	0.210	0.438	**0.175**	0.078	0.551
No ethnic homophily	0.600	0.199	0.420	0.497	0.100	**0.144**
No stereotype learning	0.550	0.193	0.519	0.399	0.052	0.602
No desperation effect	0.559	0.164	0.644	0.595	0.081	0.628
Equal access to returns	0.526	**0.015**	0.540	0.515	0.038	0.593
Equal access + no bias	0.538	0.010	0.432	0.036	0.060	0.039
Redistribution (35% tax)	**0.304**	0.138	0.652	0.598	0.000	0.626
Forced contact	0.582	0.202	0.377	0.456	0.068	0.568

That collapse looks like an extreme case of ethnic fragmentation. It is not. Figure 3b re-runs the identical world with identity bias fully intact but productivity equalised across groups. Group C's in-group cooperation recovers to 0.74, essentially group A's 0.77. The behavioural difference between the panels is generated by poverty, not by identity: scarce agents defect, their partners learn, and the group's reputation — and then its access to cooperative surplus — degrades. This is the model's central feedback loop, and it is worth emphasising that in panel (b) the agents hold exactly the same prejudices as in panel (a).

4.2 Decomposing the mechanisms

Table 1 and Figure 4 report the counterfactuals.

Result 1: identity bias alone does not open a group gap. Equalising access to returns while leaving identity bias untouched drives the between-group share from 22.3 % to 1.5 %, a 93 % reduction, even though in-group bias remains at 0.515 — statistically indistinguishable from the baseline's 0.522. Adding

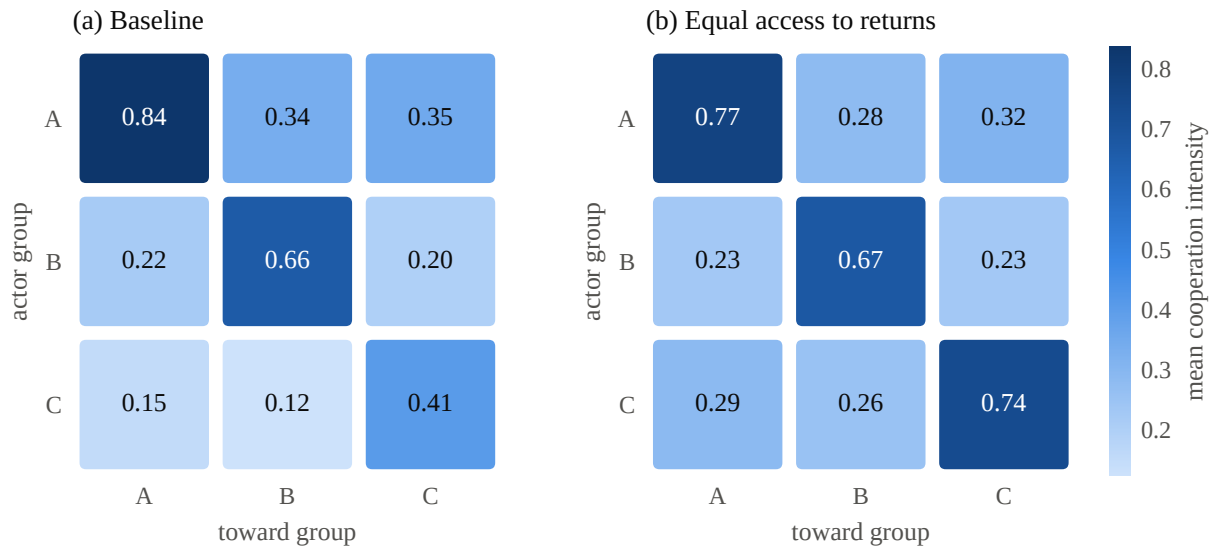


Figure 3: Mean cooperation intensity from the row group toward the column group, on a shared scale. (a) Baseline. (b) Equal access to returns, with identity bias unchanged. Group C's near-total behavioural collapse in (a) is a consequence of its poverty, not of its identity: equalising access restores its in-group cooperation to 0.74.

no bias on top of equal access moves the between-group share only from 0.015 to 0.010. Read down column two of Figure 4b: every identity-channel intervention shifts the between-group share by at most 0.029, while the access interventions shift it by 0.21.

The mechanism is straightforward once stated. When every group favours its own by the same amount, favouritism is close to a wash in aggregate. What survives is a modest majority advantage arising from group size: a member of group A meets co-ethnics half the time and enjoys the in-group bonus on half its interactions, whereas a member of group C does so only a fifth of the time. That residual is visible and real — under equal access, group A holds 52.3 % of resources on 50 % of the population and group C holds 18.8 % on 20 % — but it is an order of magnitude smaller than the productivity effect it is often invoked to explain.

We regard this as the paper's main result, and note that it cuts against a common informal reading of the diversity-and-public-goods literature (Alesina et al., 1999; Habyarimana et al., 2007). Those findings concern the *level* of cooperation in diverse settings, which our model reproduces; they do not by themselves license an inference about the *between-group distribution*, which our model suggests requires an asymmetry that symmetric favouritism does not supply.

Result 2: preference and segregation are near-orthogonal. Removing the taste for co-ethnics cuts in-group bias from 0.522 to 0.175 (a 66 % reduction) but leaves assortativity at 0.551 against a baseline 0.620. Removing ethnic homophily in tie formation collapses assortativity from 0.620 to 0.144 — and cross-group ties rise from 19.9 % to 50.1 % of all ties — while in-group bias barely moves, from 0.522 to 0.497. Each intervention almost fully addresses one outcome and almost entirely fails to address the other.

This has a policy reading. Interventions aimed at mixing (desegregating neighbourhoods, schools, workplaces) and interventions aimed at preferences (contact-based prejudice reduction, norm campaigns) are not substitutes in this model, and evaluating one against the other's metric will make it look ineffective. Our *forced contact* scenario makes the point sharply: raising the share of interactions with strangers to 80 % lowers in-group bias only modestly (0.522 → 0.456) while cutting the overall cooperation

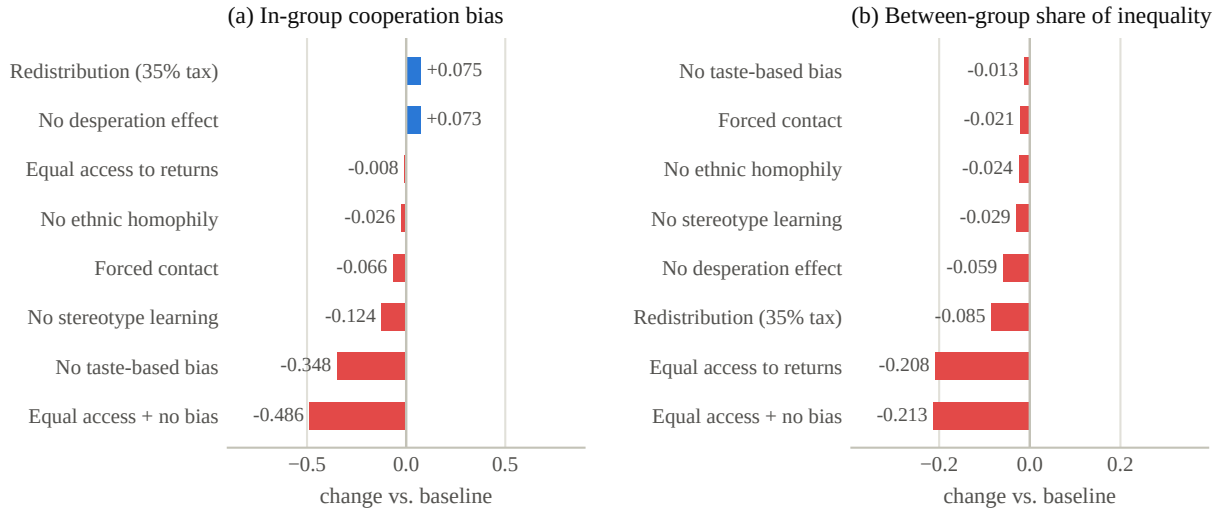


Figure 4: Change from baseline when each mechanism is disabled. (a) In-group cooperation bias: taste-based preference is the dominant term. (b) Between-group share of inequality: only the two equal-access scenarios move it materially.

rate substantially ($0.518 \rightarrow 0.377$), because strangers carry neither dyadic history nor social proximity. Exposure without preference change is expensive.

Result 3: stereotype learning amplifies, and converts. Switching off group-level learning cuts in-group bias by roughly a quarter ($0.522 \rightarrow 0.399$) but does not eliminate it. The sweep in Figure 5a shows why: bias runs from 0.11 to 0.82 across the grid, and the two parameters multiply rather than add. At $\beta_{\text{eth}} = 0$, raising λ_{grp} from 0 to 2.25 moves bias only from 0.110 to 0.176; at $\beta_{\text{eth}} = 2.4$ the same change moves it from 0.607 to 0.819. Learning compounds an existing preference into a self-confirming one; it does not manufacture one from nothing.

But the same sweep shows something we did not anticipate, and it is arguably more consequential. Along the $\beta_{\text{eth}} = 0$ column — agents with *no taste for discrimination whatsoever* — raising λ_{grp} from 0 to 2.25 raises the Gini from 0.548 to 0.613 and nearly triples the destitute share, from 5.4 % to 14.8 %. Group-level learning alone, applied to a group that is genuinely poorer and therefore genuinely more prone to scarcity-driven defection, produces group-level exclusion. The agents are not prejudiced; they are correct, on average, about the current behaviour of a disadvantaged group, and acting on that correct average generalisation makes the disadvantage worse. This is statistical discrimination (Phelps, 1972; Arrow, 1973) arising endogenously, with the self-confirming structure of Coate and Loury (1993), from a purely distributional initial condition.

Result 4: scarcity dissolves the in-group premium. The natural expectation is that hardship sharpens group boundaries. In this model it does the opposite. Figure 5b sweeps scarcity sensitivity against crisis exposure: in-group bias falls monotonically from 0.583 in the benign corner to 0.284 in the harsh one, a 51 % reduction, while the Gini rises from 0.512 to 0.667 and the destitute share from 6.8 % to 24.2 %. Desperation is indiscriminate: an agent near the floor defects on co-ethnics and strangers alike, so the in-group premium is the first thing poverty destroys. Solidarity, in this model, is a normal good.

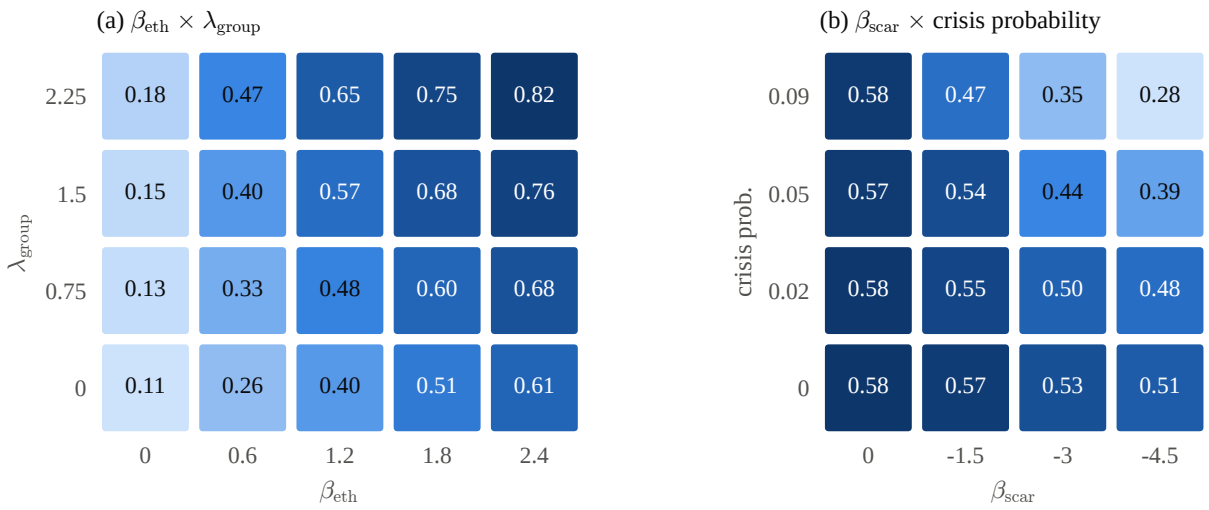


Figure 5: In-group cooperation bias across two parameter grids, 3 seeds per cell. (a) Taste-based preference β_{eth} against stereotype-learning weight λ_{grp}: the two multiply. Note the $\beta_{\text{eth}} = 0$ column, where bias stays near zero but destitution nonetheless rises with λ_{grp} (see text). (b) Scarcity sensitivity β_{scar} against crisis probability: economic stress *reduces* measured in-group bias, because desperate agents defect on co-ethnics too.

Result 5: redistribution raises cooperation. A 35 % flat tax with equal per-capita transfer cuts the Gini from 0.578 to 0.304, eliminates destitution entirely, and raises the cooperation rate from 0.518 to 0.652. The channel is (2): transfers lift agents out of the region where the scarcity term binds, removing desperation-driven defection and enlarging the cooperative surplus. Group C's resource share rises from 4.9 % to 13.4 %.

Redistribution also *raises* measured in-group bias, from 0.522 to 0.598. This is not a perverse effect but a measurement one, and it follows directly from Result 4: with indiscriminate poverty-driven defection removed, what remains visible is the underlying identity preference, which was always there. An analyst who measured in-group bias before and after such a policy and concluded that it had increased prejudice would be misreading a compositional change as a preference change.

5 Discussion

The model's assumptions connect to a contested empirical literature, and we want to be explicit about where our specification takes a side.

The most consequential assumption is $\beta_{\text{scar}} < 0$: that economic scarcity reduces willingness to cooperate. The evidence here is genuinely mixed. Field evidence from Namibian pastoralists finds that resource scarcity raises antisocial behaviour (Prediger et al., 2014), which supports our sign, and the broader psychology-of-poverty literature documents scarcity narrowing attention and shortening horizons (Shah et al., 2012; Mani et al., 2013; Haushofer and Fehr, 2014). But the cognitive-load mechanism specifically has failed to replicate in a well-powered natural experiment (Carvalho et al., 2016), and the influential finding that lower-class individuals are *more* prosocial (Piff et al., 2010) has not survived large-sample and field replication (Korndörfer et al., 2015; Andreoni et al., 2021). Our model should be read as exploring the consequences of the negative-sign assumption, not as evidence for it. The $\beta_{\text{scar}} = 0$ column of Figure 5b is, in effect, the robustness check: with desperation switched off, in-group bias is roughly flat in crisis exposure, and the Result 4 pattern disappears.

Similarly, whether poverty traps exist at all is disputed: asset-based accounts argue for genuine multiple equilibria (Carter and Barrett, 2006; Barrett et al., 2016), while Kraay and McKenzie (2014) argue the evidence for the classic mechanisms is weak. Our trap is behavioural rather than technological — it arises because destitute agents are excluded from cooperative surplus, not because of an S-shaped savings function — which we think is a mechanism the empirical debate has under-explored, but it remains an assumption.

Where the model is on firmer ground is the in-group favouritism it takes as a primitive, which is among the more robust findings in social psychology (Tajfel et al., 1971; Balliet et al., 2014), and the homophily of the interaction network (McPherson et al., 2001; Currarini et al., 2009).

Relative to prior agent-based work on wealth distribution — the Sugarscape tradition (Epstein and Axtell, 1996) and the statistical-mechanics-of-money literature (Drăgulescu and Yakovenko, 2000) — the contribution here is the coupling of a group-structured behavioural rule to the resource process in both directions, so that the distribution and the pattern of cooperation co-evolve rather than one driving the other.

6 Limitations

Several are worth stating plainly.

Symmetric bias only. Every group favours its own by the same β_{eth}. There is no group that everyone defects on, no status hierarchy, and no historically inherited animus. Given Result 1 — that symmetric favouritism cancels in aggregate — the obvious next experiment is asymmetric discrimination, which we expect to change the conclusion substantially and which we have not run.

No institutions, no coalitions, no inheritance. Agents cannot form groups, enforce norms on one another beyond bilateral reciprocity, or transmit wealth to descendants. The redistribution lever is imposed exogenously rather than chosen.

Calibrated, not fitted. Parameters were chosen for qualitative plausibility. The magnitudes reported are properties of this parameterisation. We have tried to report the directions, which hold across seeds and across the sweep grids, and to flag where a magnitude is doing rhetorical work.

Composite cooperation measure. The reported "cooperation intensity" averages a binary donation decision and a continuous trust stake, which is convenient but not a single well-defined behavioural quantity.

One population structure. All results use three groups with shares $(0.5, 0.3, 0.2)$. Group-size effects matter for Result 1, so the magnitude of the residual majority advantage will depend on the partition; we have not swept it.

7 Conclusion

We built a model designed to let ethnic favouritism generate between-group inequality, and it largely declines to. Symmetric in-group bias shapes who cooperates with whom and how segregated the network becomes, but it produces almost no durable group-level resource gap on its own; differential access to returns produces nearly all of it. What identity supplies is not the gap but the machinery for entrenching one: group-level reputation learning converts an economic disadvantage into a group-level stigma even among agents with no taste for discrimination, and that conversion works through correct-on-average generalisation rather than through prejudice.

The complementary finding is that scarcity is a solvent rather than a catalyst for group solidarity. As hardship rises, cooperation falls faster within groups than between them, and the in-group premium

erodes. A corollary that we did not expect and that has a practical edge: because poverty suppresses cooperation indiscriminately, alleviating it can *raise* measured in-group bias while making everyone better off. Measured prejudice and realised discrimination can move in opposite directions, and an evaluation that tracks only the former can draw exactly the wrong conclusion.

Reproducibility. The simulation engine, a suite of 46 verification checks (closed-form metric identities, payoff accounting, conservation and degenerate-parameter cases), the experiment runner and the figure scripts accompany this paper. All results are reproducible from a fixed seed sequence; an interactive viewer of the full result set is included.

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A Baseline parameter values

Table 2: Baseline parameterisation.

Symbol	Meaning	Value
Population		
N	agents	500
G	ethnic groups	3
(π_g)	population shares	$(0.5, 0.3, 0.2)$
(μ_g)	mean initial resources	$(12, 10, 8)$
$(\bar{a}_g)$	mean productivity	$(1.2, 1.0, 0.75)$
σ_a	productivity dispersion	0.35
K	cultural dimensions	4
Network		
$\bar{d}$	target mean degree	8
h_e, h_c, h_w	homophily weights (ethnic, cultural, wealth)	$1.5, 1.2, 0.8$
p_{loc}	probability partner drawn from network	0.75
ρ	per-agent rewiring rate	0.02
Decision rule		
β_0	baseline propensity	-0.6
β_{eth}	co-ethnic preference (centred)	1.2
β_{cult}	cultural similarity	1.0
β_{soc}	social proximity	0.6
β_{norm}	own cooperative-norm trait	1.5
β_{own}	own affluence	0.5
β_{gap}	partner-richer gap	-0.6
β_{scar}	scarcity / desperation	-2.5
η	scarcity buffer, in rounds of subsistence	5
$\lambda_{\text{grp}}, \lambda_{\text{dyad}}$	learned-belief weights	$1.0, 1.2$
$\alpha_{\text{grp}}, \alpha_{\text{dyad}}$	learning rates	$0.02, 0.30$
σ_ε	decision noise	0.25
Games		
ϕ	share of interactions that are trust games	0.35
γ, m	donation cost, multiplier	$0.6, 2.4$
$\tau, \bar{\tau}, m_\tau$	trust stake fraction, cap, multiplier	$0.25, 10, 2.5$
p_{rich}	probability wealthier agent is investor	0.8
Economy		
y_0, r, θ	flat income, return on holdings, curvature	$1.2, 0.30, 0.6$
$\bar{s}, \delta$	subsistence, depreciation	$1.6, 0.10$
σ_y	income shock dispersion	0.15
q, λ	crisis probability, severity	$0.02, 0.55$
t	tax rate (baseline)	0
Run		
	rounds; seeds per scenario	350; 5